

LAB 3 – ARCHITECTURAL PATTERN SELECTION & COMPONENT MODELLING

Blood Bank Inventory & Emergency Donor Matcher

Architecture Selection

We chose Client-Server Architecture for the Blood Bank Inventory & Emergency Donor Matcher.

Architectural Choice and Reasons

1. Centralized emergency processing: the server can validate emergency blood requests, check real-time inventory, perform ABO/Rh(D) compatibility checks, and coordinate donor matching from one controlled processing point.
2. Centralized access for different users: Emergency Requesters and Blood Bank Managers can use separate interfaces while sharing consistent request, donor, inventory, and response information maintained by the server.

Security Advantage

Centralized server-side control provides a single point for enforcing protection of sensitive donor information. The design can apply TLS 1.2+ to data in transit and AES-256 to data at rest, as required by NFR-002, while controlling access to donor and medical information.

Performance Benefit

Centralized processing allows the system to check current inventory and perform donor matching without duplicating the matching logic across clients. This supports the emergency requirement to notify compatible donors within 30 seconds and the requirement for inventory changes to be reflected within 2 seconds.